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Model predictive control-based satellite docking control for on-orbit refueling mission

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ABSTRACT

On-orbit satellite refueling is an essential aspect of satellite operations, as it enables satellites to prolong their lifetime and improves their overall performance. One of the critical challenges in the docking phase of such a mission is the fuel sloshing disturbance, which can affect the accuracy and safety of the docking process. In this study, we propose a control strategy for the docking phase of a refueling mission, where the objective is to safely and efficiently refuel a stationary target satellite. We use a combination of model predictive control and linear quadratic gaussian control to address the fuel sloshing disturbance, which is modeled using a spherical pendulum. The effectiveness and feasibility of the proposed approach are evaluated through numerical simulations using the Zero-G Lab facilities of the University of Luxembourg. The results demonstrate that the proposed strategy is capable of achieving a safe and fuel-efficient docking trajectory in the presence of fuel sloshing disturbance.

1. Introduction

Satellites are complex systems without maintenance and repair infrastructure, rendering expensive satellites useless and contributing to space debris. Extending the satellite lifespan has garnered significant interest, since it can reduce the need of replacement [1]. On-orbit satellite refueling offers a promising solution, revolutionizing space missions by prolonging lifespan, reducing launch costs, and enhancing mission flexibility [2–4]. Refueling satellites in orbit extends their operational time, benefiting communication and weather forecasting satellites in GEO [5]. This cost-effective and environmentally friendly approach reduces the need for new satellite launches and mitigates space debris concerns [6]. Moreover, on-orbit refueling enables satellite servicing, debris removal, and the construction of space-based infrastructure, including space stations and telescopes, improving space sustainability, safety, and the feasibility of long-term human presence and exploration in space [7,8]. It also has been shown that by refueling in lunar orbit and the Sun-Earth Lagrange point 2 (L2), payload mass can be substantially increased during exploration missions to Mars [9]. This technology holds immense potential for advancing space missions and ensuring the long-term viability of space activities.

The field of satellite refueling has seen significant research in recent years. Many organizations and companies have been working on developing technology and demonstrating its feasibility. One of the leading research areas has been developing new propulsion systems for on-orbit servicing missions focusing on the trade-offs between specific impulse and thrust. For example, the use of electric propulsion for on-orbit servicing and the challenges that need to be overcome are discussed in [10], where a theoretical model of electric propulsion systems is also presented that can be implemented for on-orbit servicing. Research has also been conducted on the robotic servicing technologies needed for on-orbit servicing. Many papers have been published on developing new robots and manipulator arms for performing satellite maintenance and repairs, as well as on the software and control systems needed for operating these robots [11]. A brief review of critical techniques required for orbital refueling and related projects is given in [12].

However, despite the development of the required technologies, due to the lack of operational testing, their reliability is under question. The refueling strategies and corresponding algorithms for refueling multi satellites have been investigated, such as one-to-one (O2O) refueling [13], one-to-many (O2M) refueling [13], many-to-many (M2M) refueling [5], peer-to-peer (P2P) refueling [14], egalitarian P2P refueling [15], cooperative P2P refueling [16], and mixed refueling strategies [17]. Moreover, the sequence of refueling has been defined as an optimization problem, which has been solved using standard optimization algorithms, including the auc-

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Nomenclature

b	Distance between nozzles and center of mass
D_s	Minimum safe distance to the target satellite
$\boldsymbol{\zeta}$	Measurement vector
θ	Polar angle
$\boldsymbol{F}$	State transition matrix
f_g	Erath's gravitational force
g	Gravitational constant
$\mathbf{H}$	State transition matrix
$\mathbf{I}$	Moment of inertia
$\boldsymbol{K}$	Feedback gain matrix
L	Lagrangian of the system
l	Pendulum length
M	Dry mass
m	Fuel mass
n	Orbital angular rate of the target satellite
$\boldsymbol{\xi}$	State vector
ϕ	Azimuthal angle
$\mathbf{Q}$	Covariance of the process
$\boldsymbol{\rho}$	Covariance of the observation noises
$\boldsymbol{p}$	Generalized coordinates vector
$\boldsymbol{\Sigma_p}$	Vector of the generalized forces
T	Kinetic energy
ψ	Satellite rotation angle.
t	Time
τ	Control moment
U	Potential energy
$\boldsymbol{u}$	Control input vector
$\boldsymbol{w}$	Orbital angular momentum vector

tion algorithm [18], the greedy randomized adaptive search procedure [19], and the hybrid-encoding genetic algorithm [20] in the latest studies.

While most studies have focused on refueling for multiple GEO satellites [21,22], refueling Low Earth Orbit (LEO) satellites have attracted less attention. For instance, two formation flight solutions have been proposed for refueling circular satellite constellations [23]. Lu et al. [24,25] investigated the P2P strategy for refueling the LEO constellation by taking perturbation, communication link, sun illumination, hold points for different rendezvous phases, and sensor switching into account. In addition, the scheduling problem of numerous Sun-synchronous orbit satellites refueling with multiple resupply spacecraft considering J2 perturbation has been studied in [26]. Optimized strategies for pure P2P and mixed refueling in circular constellations were investigated [17]. Similarly, the circular constellation refueling with the P2P strategy has been studied in [27] to equalize the fuel among the constellation satellites. In [28], S2M and P2P refueling strategies have been compared, and it resulted that a mixed strategy with a P2P component results in better overall fuel consumption.

However, orbital refueling cannot be implemented until rendezvous and docking are achieved. Thus, a control algorithm is required for satellites rendezvous and docking in an orbital refueling mission to guarantee safety and accuracy. Since even a tiny unwanted velocity at the docking moment can destroy both satellites, satellite docking is a high-risk scenario. Although all the proposed control methods provide a certain level of performance required for satellite docking, they cannot be employed in orbital refueling missions. The reason is that the fuel sloshing, whose dynamics model is not known precisely, significantly affects the satellite's dynamics in orbital refueling missions. Thus, a robust control scheme is essential to guarantee smooth satellite docking by considering the fuel sloshing disturbance. Moreover, a collision avoidance constraint should provide the safety of satellites.

Numerous control techniques, including sliding mode control [29,30], H_∞ control [31], model predictive control (MPC) [32–36], fuzzy control [37], reinforcement learning [37,38], and adaptive saturated control [39] have been proposed by researchers for the satellite docking problem. However, none of these algorithms have considered the influence of fuel sloshing disturbance on the docking dynamics. This paper addresses this gap and aims to propose a suitable control algorithm specifically tailored for satellite rendezvous and docking in orbital refueling missions. The objective is to achieve precise, safe, and smooth rendezvous and docking procedures, accounting for the effects of fuel sloshing disturbance.

This paper presents contributions to the field of satellite rendezvous and docking in on-orbit refueling missions. The primary focus is on developing novel strategies and techniques to enhance the efficiency and effectiveness of the docking process Firstly, it presents a novel control strategy specifically designed for the docking phase, taking into account the presence of fuel sloshing disturbance. This addresses an important aspect that has been overlooked in previous research. By considering the fuel sloshing disturbance, the proposed control strategy aims to ensure accurate and safe docking maneuvers. Secondly, the paper introduces a combination of MPC and Linear Quadratic Gaussian (LQG) control techniques. This hybrid approach allows for the generation of docking trajectories that are not only safe but also fuel-efficient. By integrating the strengths of both MPC and LQG control, the proposed strategy aims to optimize the docking process while considering fuel constraints and dynamics. Lastly, the paper presents a new model for fuel sloshing using a spherical pendulum. This model provides an accurate approximation of the fuel sloshing behavior, enabling more precise control algorithms. By improving the representation of fuel sloshing dynamics, the proposed model contributes to the overall accuracy and effectiveness of the control strategy for satellite rendezvous and docking.

The rest of the paper is organized as follows: in Section 2, the general dynamical model of the orbital refueling mission is briefly outlined, providing a foundation for the subsequent discussions. Section 3 then presents the detailed guidance and control framework, highlighting the proposed control strategy that accounts for fuel sloshing disturbance. The combination of MPC and LQG control is explained, emphasizing its role in generating safe and fuel-efficient docking trajectories. The new model for fuel sloshing using a spherical pendulum is described, demonstrating its accurate approximation capabilities. Section 4 introduces the experimental setup, based on that the system is modeled. To evaluate the performance of the proposed control approach, Section 5 presents numerical simulations and describes the experimental system, showcasing its effectiveness in achieving accurate and safe docking maneuvers while considering fuel sloshing dynamics. Finally, in Section 6, the paper concludes with some remarks summarizing the contributions and highlighting the significance of the proposed control strategy for on-orbit satellite refueling missions.

2. System description

In this chapter, the system model for the development of a control strategy for satellite refueling is described, which will be tested in the Zero-G Lab at the University of Luxembourg [40], introduced in Section 4.

The schematic diagram of the system is shown in Fig. 1, representing the mission scenario where a tanker satellite executes controlled maneuvers to dock safely with a target satellite in LEO. Both satellites are assumed to operate within the same orbital plane. The tanker carries a partially filled fuel tank, which results in fuel sloshing inside the tank. This sloshing effect can impact

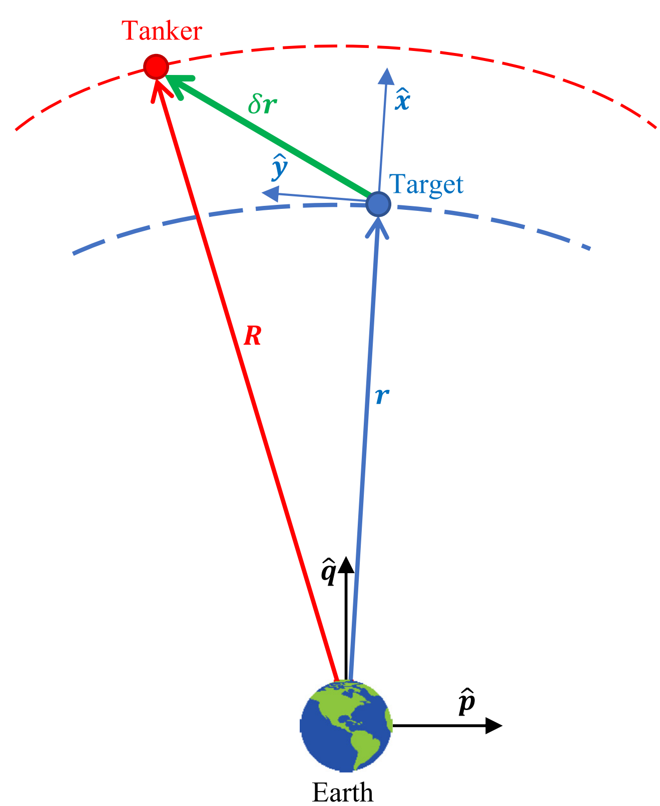

Fig. 1. The schematic diagram of the system configuration.

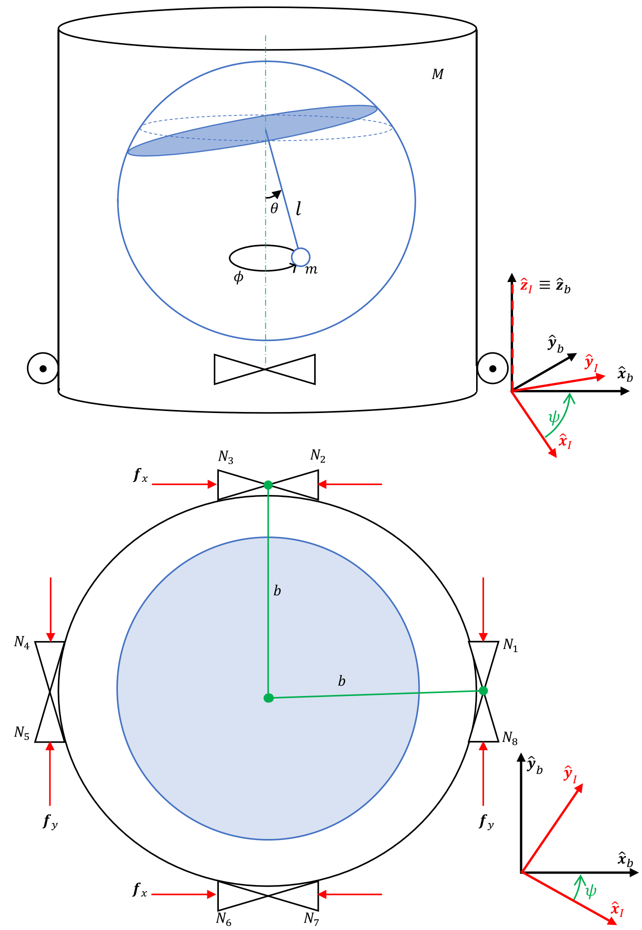

Fig. 2. The satellite model with slosh dynamics spherical pendulum analogous system.

the tanker's approach trajectory and may lead to a collision at the docking instant if not properly compensated. Throughout the paper, all vectors are treated as column matrices and are represented using bold and italic notation (e.g., $\boldsymbol{x}$). Moreover, the magnitude of a vector is represented by just italic characters without bold formatting (e.g., x).

Sloshing phenomena arise from the oscillatory motion of a liquid's free surface within a container, and it is influenced by factors such as the tank shape and the applied acceleration, including the influence of gravity. The primary mode of sloshing exhibits the most significant disturbance in the liquid's center of mass, resulting in system-wide oscillations. To analyze and understand sloshing dynamics, mechanical equivalents are often employed to describe system behavior in terms of forces and torques acting on the system. The substitution of an actual sloshing model with an equivalent oscillating model offers the advantage of simplifying motion analysis compared to the complex fluid dynamics equations.

Pendulum models have proven to be effective equivalent slosh models for simulating the motion of fluids in containers. These models possess the advantageous characteristic of frequency modulation, enabling them to adapt to varying accelerations and accurately represent a wide range of sloshing motions. Notably, the pendulum model demonstrates particular efficacy in capturing large-scale sloshing dynamics, with the accuracy of representation depending on the specific geometric attributes of the container [45]. Empirical evidence, as presented in [46], validates the behavior of the equivalent spherical pendulum model by aligning with experimental observations of stable sloshing in cylindrical tanks. Consequently, the spherical pendulum serves as an appropriate mechanical model for effectively representing sloshing phenomena.

Consider a rigid spacecraft, referred to as the tanker, operating within the orbital plane of the target satellite. The tanker possesses a spherical fuel tank and exhibits slosh dynamics that can be analogously represented by a spherical pendulum model of length l. Fig. 2 illustrates the configuration of the tanker model in both the body frame $(\boldsymbol{xyz})_b$ and the inertial frame $(\boldsymbol{xyz})_I$. The $\boldsymbol{z}$-axis, denoted as $\hat{\boldsymbol{z}}$, is assumed to be consistently aligned with the orbit angular momentum vector ($\hat{\boldsymbol{w}}$), resulting in the tanker's restricted planar movement and single-axis rotation around $\hat{\boldsymbol{z}}$, denoted as ψ.

The dry mass of the tanker, which includes the structural components, is denoted as M, while the mass equivalent of the fuel alone is represented by m. The free surface of the liquid within the fuel tank is characterized by the polar angle, θ, and the azimuthal angle, ϕ, measured with respect to the $\boldsymbol{x}_b$-axis in the body frame. The tanker is equipped with eight on/off thrusters, which are strategically positioned at a distance of b from the center of mass. These thrusters enable the generation of torque required for attitude control around $\hat{\boldsymbol{z}}$ and facilitate planar motion control within the $\boldsymbol{xy}$ plane. It is important to note that the $\boldsymbol{xy}$ plane, representing the horizontal plane of motion, is precisely aligned with the orbital plane.

This model is chosen for its computational simplicity and efficiency, which makes it suitable for real-time control applications. However, it represents a simplified approximation of sloshing behavior and may not fully capture the fluid dynamics of non-spherical tanks. Future studies could extend this approach to more complex tank geometries using advanced modeling techniques, enhancing the accuracy and robustness of the control system.

The assumptions considered for deriving the equations of motion are as follows:

1. The satellite body is assumed to be rigid and non-flexible: This assumption ensures that the structural deformations of the satellite body do not influence its motion dynamics. It simplifies the mathematical model by treating the satellite as a rigid body, which is a common assumption in spacecraft dynamics.
2. The system specifications are considered to be fixed and unchanged: The mass, inertia, thruster configuration, and other physical properties of the system remain constant throughout the experiment and simulation. This assumption ensures consistency in system behavior and avoids additional complexities introduced by time-dependent parameter variations, such as fuel depletion effects.
3. Friction between the fuel and the tank is neglected, resulting in independent rotational motion of the satellite from the pendulum rotary motion: This simplification allows for a clearer analysis of fuel sloshing dynamics by assuming that the fluid motion inside the tank does not introduce additional damping forces due to friction. This assumption is reasonable for low-viscosity fluids in space-like conditions.
4. The tanker and the target satellite are assumed to be in a single orbital plane: This assumption simplifies the relative motion equations by reducing the problem to two-dimensional in-plane dynamics. It is a valid assumption for many real-world docking scenarios where the approach is constrained to a single orbital plane.
5. The position and attitude of the target satellite are known: This ensures that the control algorithm has access to accurate state information for the target satellite, allowing for precise guidance and docking maneuvers. In real missions, this information is typically obtained from navigation sensors.
6. The position, attitude, and their time derivatives of the tanker are measurable: This assumption ensures that the state estimation system, such as an onboard Kalman filter, has sufficient observability to accurately track the tanker's motion, a necessary condition for implementing the control strategy.
7. Attitude disturbances, including gravity gradient effects, are ignored: The experimental setup does not replicate real orbital perturbations such as gravity gradients, third-body influences, or solar radiation pressure. Since the control system is tested in a controlled laboratory environment, these external effects are not considered in the modeling phase.
8. The fuel remains in a stable condition without instability: The sloshing model assumes that the liquid remains within predictable oscillatory motion and does not exhibit chaotic behavior, such as violent splashing or cavitation. This ensures that the spherical pendulum approximation remains valid throughout the docking maneuver.

2.1. System dynamics

The equation of motion of the tanker in the inertial reference frame can be obtained from Lagrange equations as follows [47]

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\boldsymbol{p}}} - \frac{\partial L}{\partial \boldsymbol{p}} = \boldsymbol{\Sigma_p} \tag{1}$$

where $L = T - U$ is the Lagrangian of the system, $\boldsymbol{p} = [x_I\, y_I\, \theta\, \phi\, \psi]^{\mathrm{T}}$ denotes the generalized coordinates vector of the system, $\boldsymbol{\Sigma_p}$ is the vector of the generalized forces acting on the system, here it is the summation of the control force/moments and the earth gravitational forces. Moreover, T and U are the kinetic energy and the potential energy of the system.

Assuming that the pendulum is connected to the satellite body's center of mass, the kinetic energy of the system is given by

$$\begin{aligned} T = \tfrac{1}{2}m\Big[&l^2\dot{\theta}^2 + l^2\sin^2\theta\dot{\phi}^2 + 2l\big(\dot{x}_I\cos\theta\cos\phi + \dot{y}_I\cos\theta\sin\phi\big)\dot{\theta} \\ &+2l\big(\dot{y}_I\sin\theta\cos\phi - \dot{x}_I\sin\theta\sin\phi\big)\dot{\phi}\Big] \\ &+\tfrac{1}{2}(m+M)\big[\dot{x}_I^2 + \dot{y}_I^2\big] + \tfrac{1}{2}I\dot{\psi}^2 \end{aligned} \tag{2}$$

where $\mathbf{I}$ represents the satellite moment of inertia. The potential energy due to the fuel sloshing can be expressed as

$$U = -mgl\cos\theta \tag{3}$$

where g represents the gravitational constant. Substituting Eqs. (2) and (3) in (1), the equations of motion of the system in the inertial reference frame can be achieved as follows

$$\begin{aligned} &(m+M)\ddot{x}_I = f_x + f_{g_x} + ml\Big(\sin\theta\cos\phi\Big(\dot{\theta}^2+\dot{\phi}^2\Big) + 2\cos\theta\sin\phi\dot{\theta}\dot{\phi}\Big) \\ &\quad -ml\Big(\cos\theta\cos\phi\ddot{\theta} - \sin\theta\sin\phi\ddot{\phi}\Big) \\ &(m+M)\ddot{y}_I = f_y + f_{g_y} + ml\Big(\sin\theta\sin\phi\Big(\dot{\theta}^2+\dot{\phi}^2\Big) - 2\cos\theta\cos\phi\dot{\theta}\dot{\phi}\Big) \\ &\quad -ml\Big(\cos\theta\sin\phi\ddot{\theta} + \sin\theta\cos\phi\ddot{\phi}\Big) \\ &l\ddot{\theta} = l\sin\theta\cos\theta\dot{\phi}^2 - g\sin\theta - \big(\ddot{x}_I\cos\theta\cos\phi + \ddot{y}_I\cos\theta\sin\phi\big) \\ &l\sin\theta\ddot{\phi} = \ddot{x}_I\sin\phi - \ddot{y}_I\cos\phi - 2l\cos\theta\dot{\theta}\dot{\phi} \\ &\mathbf{I}\ddot{\psi} = \tau \end{aligned} \tag{4}$$

where $\boldsymbol{f} = [f_x\, f_y]^{\mathrm{T}}$ and τ are the control force and moment generated by thrusters and $\boldsymbol{f}_g = [f_{g_x}\;\, f_{g_y}]^{\mathrm{T}}$ represent the Erath's gravitational force vector acting on the tanker. The detail information about calculating Erath's gravitational force can be found in [48].

Indeed, it can be argued that while the tanker is assumed to have a planar movement and single-axis rotation around the z-axis, the fuel sloshing is modeled as a 3-dimensional movement using a spherical pendulum with two angles, θ and ϕ. This may appear contradictory at first. However, it should be noted that in the Zero-G Lab, only planar motions can be effectively simulated. Therefore, the out-of-plane movements of the pendulum, which could potentially cause out-of-plane motion of the tanker, are effectively limited within the simulated planar motion. As a result, the assumption of planar movements for the tanker is justified within the context of the Zero-G Lab's simulation capabilities.

Let consider the spacecraft's nonlinear equations of motion as

$$\dot{\boldsymbol{\xi}}(t) = \boldsymbol{f}\big(\boldsymbol{\xi}(t), \boldsymbol{u}(t)\big) \tag{5}$$

where $\boldsymbol{\xi} = [x_I\; \dot{x}_I\; y_I\; \dot{y}_I\; \theta\; \dot{\theta}\; \phi\; \dot{\phi}\; \psi\; \dot{\psi}]^{\mathrm{T}}$ denotes the system state vector, $\boldsymbol{u} = [f_{x_I}\;\, f_{y_I}\;\, \tau]^{\mathrm{T}}$ represents the control input, and $\boldsymbol{f}(\cdot,\cdot) : (\mathbb{R}^n, \mathbb{R}^m\,) \to \mathbb{R}^n$ denotes a nonlinear function of the system. n and m are the number of system states and control inputs, respectively. It should be noted that the system measurement vector is $\boldsymbol{\zeta} = [x_I\, y_I\; \psi]^{\mathrm{T}}$.

Let $\boldsymbol{\xi}'(t^+)$ denote the system state in a finite upcoming time horizon, obtained by applying $\boldsymbol{u}(t) = \boldsymbol{u}(t^-)$ to (5) having the initial condition $\boldsymbol{\xi}'(t) = \boldsymbol{\xi}(t)$, where $\boldsymbol{u}(t^-)$ corresponds to the most recently applied control command to the system prior to time t. Using the first order approximation of the system (5) around $\boldsymbol{\xi}'(t)$ and $\boldsymbol{u}(t^-)$, the resulting Linear Time-Varying (LTV) system is as

$$\dot{\boldsymbol{\xi}}(t) = \mathbf{A}_t\boldsymbol{\xi}(t) + \mathbf{B}_t\boldsymbol{u}(t) + \boldsymbol{d}(t) \tag{6}$$

where $\mathbf{A}_t \in \mathbb{R}^{n\times n}$ and $\mathbf{B}_t \in \mathbb{R}^{n\times m}$ are defined as follows

$$\mathbf{A}_t = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{\xi}}\big|_{\boldsymbol{\xi}'(t),\boldsymbol{u}_0},\;\; \mathbf{B}_t = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\big|_{\boldsymbol{\xi}'(t),\boldsymbol{u}_0} \tag{7}$$

and $\boldsymbol{d}(t) \in \mathbb{R}^{n\times 1}$ is defined as

$$\boldsymbol{d}(t) = \boldsymbol{\xi}'\,(t) - \mathbf{A}_t\boldsymbol{\xi}'(t) - \mathbf{B}_t\boldsymbol{u}\big(t^-\big). \tag{8}$$

The LTV system (8) can be utilized over a finite time horizon to approximate the nonlinear system (5) for further use. The $\mathbf{A}_t$ and

$\mathbf{B}_t$ matrices can be derived in explicit forms as follows:

$$\mathbf{A}_t = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & a_{25} & a_{26} & a_{27} & a_{28} & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & a_{45} & a_{46} & a_{47} & a_{48} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ a_{61} & 0 & a_{63} & 0 & a_{65} & a_{66} & a_{67} & a_{68} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ a_{81} & 0 & a_{83} & 0 & a_{85} & a_{86} & a_{87} & a_{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \text{ and } \mathbf{B}_t = \begin{bmatrix} 0 & 0 & 0 \\ \frac{1}{m+M} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{1}{m+M} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{I} \end{bmatrix},$$

where

$$a_{25} = \frac{ml}{m+M}\left(\cos\theta\cos\phi\left(\dot{\theta}^2 + \dot{\phi}^2\right) - 2\sin\theta sin\phi\ \dot{\theta}\dot{\phi}\right),$$

$$a_{26} = \frac{2ml}{m+M}\left(\sin\theta\cos\phi\dot{\theta} + \cos\theta sin\phi\ \dot{\phi}\right),$$

$$a_{27} = \frac{ml}{m+M}\left(-\sin\theta\sin\phi\left(\dot{\theta}^2 + \dot{\phi}^2\right) - 2\cos\theta cos\phi\ \dot{\theta}\dot{\phi}\right),$$

$$a_{28} = \frac{2ml}{m+M}\left(\sin\theta\cos\phi\dot{\phi} + \cos\theta sin\phi\ \dot{\theta}\right),$$

$$a_{45} = \frac{ml}{m+M}\left(\cos\theta\cos\phi\left(\dot{\theta}^2 + \dot{\phi}^2\right) + 2\sin\theta\cos\phi\dot{\theta}\dot{\phi}\right),$$

$$a_{46} = \frac{2ml}{m+M}\left(\sin\theta\sin\phi\dot{\theta} + \cos\theta\cos\phi\dot{\phi}\right),$$

$$a_{47} = \frac{ml}{m+M}\left(\sin\theta\cos\phi\left(\dot{\theta}^2 + \dot{\phi}^2\right) - 2\cos\theta\sin\phi\dot{\theta}\dot{\phi}\right),$$

$$a_{48} = \frac{2ml}{m+M}\left(\sin\theta\sin\phi\dot{\phi} - \cos\theta\cos\phi\dot{\theta}\right),$$

$$a_{61} = -\frac{\cos\theta\cos\phi}{l},$$

$$a_{63} = -\frac{\cos\theta\sin\phi}{l},$$

$$a_{65} = \frac{\sin\theta\cos\theta\dot{\phi}^2 - g\cos\theta}{l},$$

$$a_{67} = \frac{x_I\sin\theta\sin\phi - y_I\sin\theta\cos\phi}{l},$$

$$a_{68} = \frac{2\sin\theta\cos\theta\dot{\phi}}{l},$$

$$a_{81} = \frac{\sin\phi}{l\sin\theta},$$

$$a_{83} = -\frac{\cos\phi}{l\sin\theta},$$

$$a_{85} = -\frac{\cos\theta\dot{\theta}\dot{\phi}}{\sin\theta},$$

$$a_{86} = -\frac{2\cos\theta\dot{\phi}}{l\sin\theta},$$

$$a_{87} = \frac{x_I\cos\phi + y_I\sin\phi}{l\sin\theta}, \text{ and } a_{88} = -\frac{2\cos\theta\dot{\theta}}{l\sin\theta}.$$

3. Control scheme

As noted earlier, the control scheme should provide satellites' safety by considering collision avoidance constraints. In addition, the sloshing disturbance may cause an unwanted velocity at the docking instance that should be eliminated by the control scheme. A control scheme is proposed in this section for the docking scenario with a stationary satellite considering the fuel sloshing. The schematic diagram of the control scheme is shown in Fig. 3, where it can be seen that the scheme is formed of a path planning part, a control part, and a filtering part.

The main objective of the path planning part is to calculate a safe and accurate reference trajectory for the tanker to approach and dock with the target satellite. In addition, it should be ensured that the tanker consumes as minimum fuel as possible for the mission. The ability to predict the system's future performance by considering various types of constraints, robustness, and optimality are among the MPC features that make it an appropriate tool for generating the collision-free reference trajectory [49]. The performance of MPC has been proven in different aerospace applications including space tethered system control [50,51], asteroid

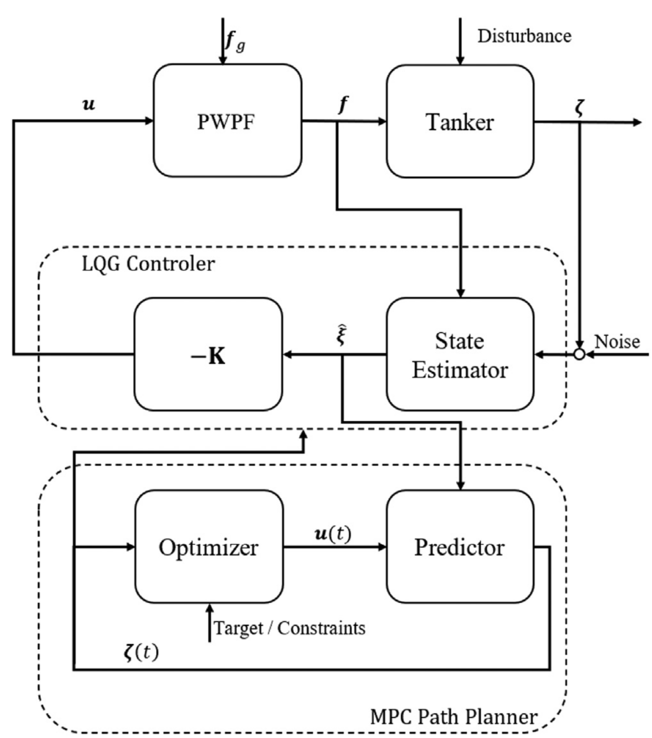

Fig. 3. The schematic diagram of the motion planning framework.

landing [52,53] and hovering [54], autonomous satellite maneuvering [55], and path planning [56].

MPC is a type of advanced control algorithm that uses a mathematical model of a system to make predictions about its future behavior. It then uses these predictions to optimize control actions in real-time to achieve a desired set point or performance. In general, MPC refers to a family of controllers which calculate the control effort by solving a finite-time optimization problem to minimize a cost function subject to constraints. An optimization problem over a finite time is solved at each sampling time using future prediction of the system's performance to determine the optimal control sequence. The basic steps of an MPC algorithm are as follows:

1. Modeling: The first step in implementing an MPC is to develop a mathematical model of the system to be controlled. This model should accurately represent the dynamics and constraints of the system.
2. Prediction: The MPC algorithm predicts the system's future behavior based on current states and inputs using the system model.
3. Optimization: The MPC algorithm then uses these predictions to optimize a performance criterion, such as minimizing an error or cost function, subject to constraints on the system.
4. Control Action: Based on the optimized solution, the MPC algorithm generates a control action to be applied to the system.
5. Measurement: After the control action is applied, the system's state is measured and fed back into the MPC algorithm.
6. Iteration: The process of predicting, optimizing, and controlling action is repeated at a regular time interval, such as every few milliseconds, to ensure that the system remains on track to meet the desired performance criteria.

It is worth noting that the MPC algorithm can be implemented in different ways, and the implementation details may vary depending on the specific application. Detailed practical information on MPC can be found in [57]. The optimization problem for the path-planning problem that should be solved at each time step is as follows:

$$\begin{aligned}
&\min_{\boldsymbol{u}(t)} \int_{t=t_0}^{t_f} \|\boldsymbol{\zeta}_{ref}(t)-\boldsymbol{\zeta}_f\|_{\boldsymbol{\Omega}}^2+\|\boldsymbol{u}(t)\|_{\omega}^2 dt\\
&s.t.\ \dot{\boldsymbol{\xi}}(t)=\mathbf{A}_t\boldsymbol{\xi}(t)+\mathbf{B}_t\boldsymbol{u}(t)+\boldsymbol{d}(t)\\
&\quad\ \ \boldsymbol{\zeta}(t)\in\boldsymbol{\Xi}\\
&\quad\ \ \boldsymbol{u}(t)\in\boldsymbol{\upsilon}\\
&\quad\ \ \boldsymbol{\xi}'(t_0)=\boldsymbol{\xi}(t_0)
\end{aligned} \tag{9}$$

where t_f denotes the final docking time, t_0 denotes the current time instant, $\boldsymbol{\zeta}_{ref}$ and $\boldsymbol{\zeta}_f$ represent the reference and desired final docking states, respectively, and $\|\ .\|_{\boldsymbol{\Omega}}^2$ denotes the weighted norm of " $\cdot$ " defined by $(.)^{\mathrm{T}}\boldsymbol{\Omega}(.)$ with $\boldsymbol{\Omega}$ being a positive definite matrix. The system is also subjected to state constraint, $\Xi\in\mathbb{R}^n$, and control input constraint, $\boldsymbol{\upsilon}\in\mathbb{R}^m$. The control framework incorporates safety constraints (Ξ) to ensure that the tanker avoids collisions with both the lab walls (which define the admissible zone to do the docking maneuver) and the target satellite during the docking maneuver. To prevent contact with the lab walls, the tanker's position is constrained within the allowable workspace of the Zero-G Lab, defined by the maximum permissible coordinates along the x and y axes. Specifically, the constraints are expressed as $|x_l|\le \mathrm{x}_{\max}$ and $|y_l|\le \mathrm{y}_{\max}$, ensuring that the tanker remains within the designated boundaries. Additionally, to maintain a safe distance from the target satellite, the constraint $x_l^2+y_l^2\ge D_s^2$ is applied, where D_s represents the minimum safe distance that must be maintained between the tanker and the target satellite. By incorporating these constraints into the MPC optimization, the control system ensures that the tanker approaches the docking port while avoiding both the lab walls and the target satellite throughout the maneuver. In addition, the input constraint $\boldsymbol{\upsilon}$ limits the maximum thrust that can be generated by the thrusters.

Given that the optimization problem (9) is a convex quadratic programming problem, it can be efficiently solved using standard Quadratic Programming (QP) solvers. The output of the MPC path planner is the reference trajectory, $\boldsymbol{\zeta_r}$.

Once the reference trajectory has been computed, the next step is to design a suitable controller for the satellite to track this trajectory while mitigating the disruptive effects of sloshing. The LQG control problem is a well-established optimal control approach known for its robust performance, even in the presence of unmodeled dynamics and systems with multiple states [58]. This control methodology combines the Kalman filter, which handles estimation tasks, with the Linear Quadratic Regulator (LQR), which addresses control objectives. By integrating these components, the LQG control algorithm becomes an effective tool capable of simultaneously addressing control and estimation challenges. It exhibits versatility in handling diverse system dynamics, including time delays and non-minimum phase systems.

It is important to note that the LQG controller requires a linear set of equations of motion, which can be achieved by assuming small deviations from the equilibrium point along the reference trajectory. Furthermore, certain states, such as the free surface angle (θ), may not be directly measurable and thus require estimation. To tackle this, an Extended Kalman Filter (EKF) [59] is employed as a state estimator. The EKF not only aids in rejecting disturbances but also facilitates the estimation of unmeasured state variables that are crucial for the controller's operation. The necessity of a state estimator arises from the significant disturbance caused by fuel sloshing motion, especially during the docking phase, where the prevention of unwanted movements and potential collisions between satellites is of utmost importance.

Let $\hat{\boldsymbol{\xi}}$ and $\hat{\boldsymbol{\zeta}}$ denote the estimated system states and system output, respectively. The LQG controller as the solution to the LQG control problem can be specified as

$$\boldsymbol{u}(t)=-\boldsymbol{K}(t)\hat{\boldsymbol{\xi}}(t) \tag{10}$$

where $\boldsymbol{K}(t)$ is the feedback gain matrix obtained from the solution of the Riccati differential equation [60]. The states can also be estimated by the EKF formulation as [60]

$$\begin{aligned}
&\dot{\hat{\boldsymbol{\xi}}}(t)=\boldsymbol{f}\left(\hat{\boldsymbol{\xi}}(t),\boldsymbol{u}(t)\right)+\boldsymbol{L}(t)\left(\boldsymbol{\zeta}(t)-\hat{\boldsymbol{\zeta}}(t)\right)\\
&\dot{\boldsymbol{P}}(t)=\boldsymbol{F}(t)\boldsymbol{P}(t)+\boldsymbol{P}(t)\boldsymbol{F}(t)^{\mathrm{T}}-\boldsymbol{L}(t)\mathbf{H}(t)\boldsymbol{P}(t)+\mathbf{Q}(t)\\
&\boldsymbol{L}(t)=\boldsymbol{P}(t)\mathbf{H}(t)^{\mathrm{T}}\boldsymbol{\rho}(t)^{-1}
\end{aligned} \tag{11}$$

in which $\mathbf{Q}(t)$ and $\boldsymbol{\rho}(t)$ represent the covariance of the process and observation noises. In addition, the state transition matrix $\boldsymbol{F}(t)$ and observation matrix $\mathbf{H}(t)$ are defined using the first-order approximation of the system (5) (similar to $\boldsymbol{A}_t$ and $\boldsymbol{B}_t$) around $\hat{\boldsymbol{\xi}}$ as follows

$$\boldsymbol{F}(t)=\frac{\partial\boldsymbol{f}}{\partial\boldsymbol{\xi}}\Big|_{\hat{\boldsymbol{\xi}}(t),\boldsymbol{u}(t)},\ \ \mathbf{H}(t)=\frac{\partial\boldsymbol{h}}{\partial\boldsymbol{\xi}}\Big|_{\hat{\boldsymbol{\xi}}(t)}\ . \tag{12}$$

To effectively mitigate the disturbances caused by fuel sloshing during the docking maneuver, the sloshing dynamics model is explicitly integrated into both the MPC and LQG control frameworks. Unlike previous studies that only use sloshing models for verification, our approach actively incorporates sloshing states into the control law to ensure real-time compensation.

The sloshing dynamics, modeled as a spherical pendulum, are included in the system equations governing the motion of the satellite. Specifically, the sloshing states (pendulum angles and their time derivatives) are incorporated into the state-space representation used in MPC for trajectory prediction and in LQG for optimal state estimation and control.

For MPC, the prediction model considers the influence of fuel sloshing on the satellite's motion, ensuring that the generated reference trajectory accounts for its effects. The control optimization explicitly includes sloshing-related constraints to minimize undesired oscillations and improve docking accuracy.

For LQG, the Kalman filter estimates the sloshing states, which are then fed into the LQR to generate corrective control inputs. By accounting for sloshing effects in the estimation and control process, LQG ensures that unexpected fuel movement does not destabilize the docking maneuver.

By integrating fuel sloshing dynamics directly into the control equations, the proposed method provides a more accurate and robust docking control strategy than previous approaches that rely on post-simulation verification [61]. This enhancement ensures that the docking trajectory remains safe, efficient, and resistant to disturbances caused by fuel motion.

As mentioned earlier, the satellite is assumed to be equipped with on/off thrusters. Thus, the continuous command control calculated by the LQG is translated to an on/off signal using the Pulse-Width Pulse-Frequency (PWPF) modulator, which is formed of a Schmidt trigger and a first-order filter. The PWPF modulator adjusts the width and frequency of pulses in the signal. The width of the pulses is controlled by adjusting the pulses' duty cycle, while the frequency of the pulses is controlled by adjusting the pulse train frequency. By adjusting both the width and frequency of the pulses, the PWPF modulator can control the signal's amplitude. The main advantages of the PWPF modulator with respect to classical on/off controllers are fuel consumption reduction and high accuracy [62]. Additional detailed information about the structure and tuning of the PWPF modulator can be found in [63].

In addition, in an actual orbital environment, a satellite experiences gravitational forces that influence its motion. However, in the Zero-G Lab, gravitational acceleration is present, and the physical constraints of the test setup do not inherently simulate the free-fall conditions of space. To compensate for this and more accurately represent orbital dynamics, the gravitational force ($\boldsymbol{f}_g$) acting on the satellite in orbit is computed and added to the control signal applied to the system via the thrusters.

This computed force is then incorporated into the control input as an additional term to ensure that the system responds similarly to how it would in an orbital scenario. The thruster actuations are adjusted accordingly to simulate the influence of gravity, ensuring that the experimental setup better represents the in-orbit dynamics of the satellite. By implementing this approach, the experimental results from the Zero-G Lab can be more accurately extrapolated to real orbital conditions, bridging the gap between ground-based testing and space operations.

4. Experimental setup

The Zero-G Lab features an advanced mechatronic system known as a floating platform, Fig. 4, designed to simulate the microgravity environment of space and enable the testing and validation of various space technologies and systems. The platform, developed through collaboration between different groups of the university including SpaceR, SpaSys, CVI2, and ARG, will serve as the experimental setup for testing the control strategy for satellite refueling.

The Zero-G Lab features a spacious experimental room measuring 5 m × 3 m × 2.3 m, equipped with two floating platforms designed to move with minimal friction over a precisely leveled epoxy floor. In the context of the Zero-G Lab, the lab floor serves as the reference plane for simulating microgravity conditions and conducting experiments related to satellite refueling. Each platform

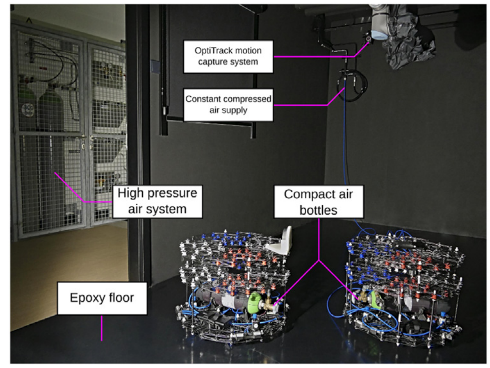

Fig. 4. Two floating platforms operating in Zero-G Lab [44].

is equipped with air bearings that release high-pressure air toward the floor, creating an air cushion that eliminates mechanical contact and simulates microgravity conditions.

The platforms are actuated by eight nozzles, capable of generating forces up to 1 N under a pressure of 10 bar. The specific configuration and arrangement of these nozzles are detailed in [41]. The platform's position, velocity, orientation, and angular velocity are tracked in real-time using six OptiTrack Prime 13 W cameras mounted within the lab, operating at 240 Hz. An active marker positioned at the center of the platform's top plate ensures accurate motion capture and state estimation. The system integrates seamlessly into the ROS (Robot Operating System) network, with a ROS-MATLAB bridge enabling real-time control and data processing using MATLAB, facilitating both experimentation and performance evaluation [42].

While the Zero-G Lab does not provide a perfect replication of the microgravity environment of space, it serves as a valuable testbed for approximating the translational and rotational dynamics encountered in orbit. The floating platforms utilized in the lab enable near-frictionless planar motion, effectively simulating the two-dimensional dynamics of spacecraft maneuvering in orbit. Additionally, the rotational dynamics of the satellite are replicated through controlled torque application, allowing us to analyze the impact of fuel sloshing on attitude stability.

It is important to acknowledge the limitations of this experimental setup. Unlike an actual orbital environment, the Zero-G Lab does not account for factors such as gravity gradient effects, third-body perturbations, and atmospheric drag. Furthermore, while our study considers a spherical pendulum model to represent fuel sloshing, the real fluid dynamics in orbit can be influenced by additional microgravity effects that are not entirely captured in our lab-based simulations. Despite these limitations, the experimental setup remains a useful platform for validating the core principles of our proposed control approach, as it allows us to test and refine algorithms under controlled conditions before their potential application in actual space missions.

It is assumed that the necessary orbital maneuvers [43] to align the tanker with the orbital plane of the target satellite are first performed. Subsequently, the precise planar rendezvous and docking maneuvers are to be executed. By focusing on this critical aspect of satellite refueling, the aim is to develop a control strategy that can effectively address the challenges associated with the precise rendezvous and docking process in the simulated microgravity environment provided by the Zero-G Lab.

5. Results and discussions

In this section, the performance of the proposed control approach is investigated using numerical simulations. First, the pro-

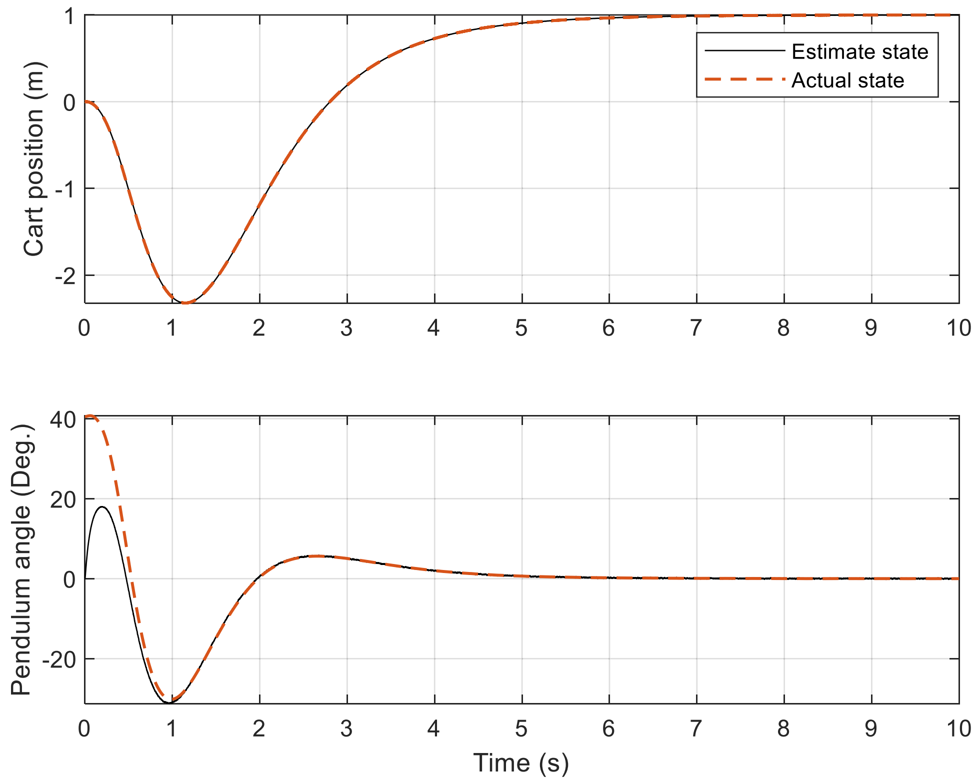

Fig. 5. The output response of the inverted pendulum control problem using the proposed control method.

posed control method is applied to the inverted pendulum control problem as a benchmark, finally it is applied to the fuel sloshing problem. Notably, the variable-step fourth order Runge-Kutta algorithm was implemented for numerical integration within a MATLAB computational environment. Additionally, the Sequential Quadratic Programming (SQP) algorithm was utilized to solve the optimization problem.

5.1. Inverted pendulum control

In this section, we present the simulation results focusing on the performance of the system with the estimated pendulum angle obtained using the proposed control combination. The system under study considers the output as the position and velocity of the cart, while the pendulum angle and its time derivative are estimated. The primary objective is to assess the controllability and stability of the inverted pendulum system. The dynamic equation for the inverted pendulum system utilized in this study is based on the reference [64].

To generate the reference trajectory, MPC is employed with a prediction horizon of 5 s and a time step of 0.1 s. The MPC algorithm aims to determine the optimal control inputs to achieve the desired trajectory. Furthermore, a 4x4 diagonal matrix $\boldsymbol{\Omega}$ is utilized, with its diagonal elements set to 1 for position and angle-related variables, and 10 for the time derivative-related variables. Additionally, ω is assigned a value of 10.

For the simulation, the initial conditions involve commanding the cart to move 1 m, while the initial estimation of the pendulum angle has an error of 40°. These initial conditions provide a starting point for evaluating the system's behavior and performance.

The system response is shown in the Fig. 5, indicating the convergence of the estimated angle to the actual value approximately one second after the start. The regulation of the pendulum angle is also observed. The initial under shoot of the cart is attributed to the initial error in estimating the pendulum angle. Furthermore, the controlled movement of the cart towards the desired position is evident, demonstrating smooth control.

Disabling the angle estimation process and relying solely on cart position data for control in the inverted pendulum system resulted in divergence and instability. Comparing this configuration with accurate angle estimation using the EKF highlighted the pivotal role of angle estimation in achieving controllability and stabilization with minimal overshoot. Incorporating angle estimation techniques, such as the EKF, in control algorithms for the inverted pendulum system is crucial for ensuring stability and controllability. A robustness analysis demonstrated the resilience and effectiveness of the proposed control algorithm with angle estimation in compensating for external disturbances and parameter variations, maintaining stability throughout the system's operation.

5.2. Fuel sloshing control

A LEO satellite in a circular sun-synchronous 600 km orbit is the target satellite for refueling. The tanker is assumed to be placed five meters behind the target satellite in the same orbit for docking. The system's initial configuration is shown in Fig. 6, where blue and green colors denote the tanker and target satellites, respectively. In addition, an admissible zone is considered for the docking maneuver, which is indicated in Fig. 6 by black borders. The tanker is equipped with eight 1-N nozzles. The rest of the tanker's model characteristics are listed in Table 1.

It should be noted that the scenario parameters are selected based on the Zero-G Lab facilities of the University of Luxembourg [65], such that the system can be implemented and validated in the laboratory for further studies. MPC with a prediction horizon of 100 s and a time step of 1 s is employed to generate the reference trajectory. Moreover, $\boldsymbol{\Omega}$ is a 6 × 6 diagonal matrix whose position- and angle-related diagonal elements are 1, and the time derivative-related elements are 100. In addition, $\boldsymbol{\omega}$ refers to a 3 × 3 diagonal matrix with all diagonal elements equal to 1000. Further-

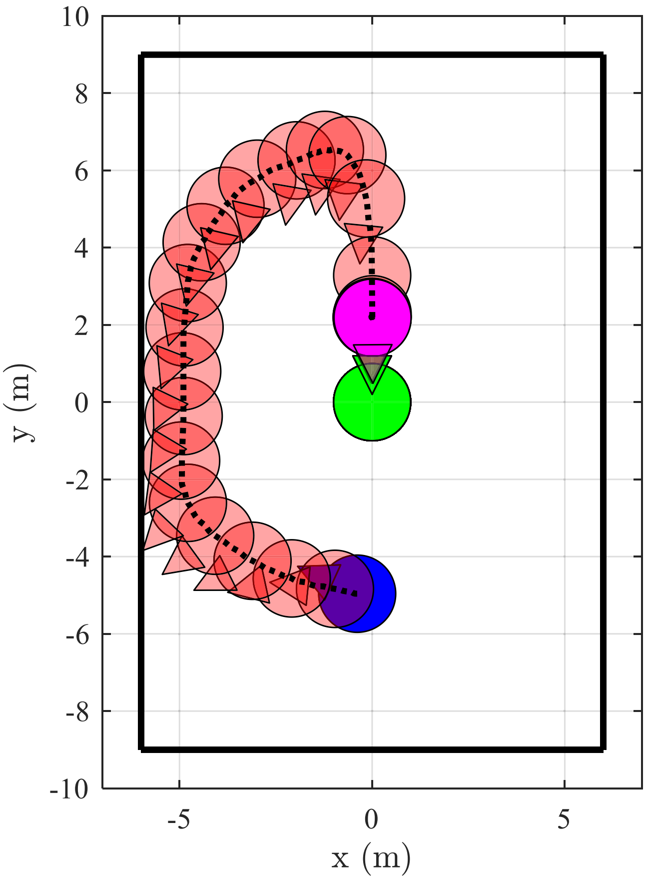

Fig. 6. The docking trajectory in the RSW reference frame.

Table 1
The tanker's model physical characteristics.

Parameter	Value	Unit
m	5	Kg
M	5	kg
l	0.4	m
b	1	m
$\mathbf{I}$	4.5	Kg·m^2

more, the observation noise covariance $\boldsymbol{\rho}$ is chosen based on the accuracy of the motion capture system as described in [66].

The controller gains are tuned to ensure that the position error remains within 10 cm of the reference trajectory throughout the docking maneuver, while the docking accuracy at the final approach is maintained within 3 cm, considering the size of the docking port. Additionally, the velocity at the docking instant is constrained to be <1 cm/s to ensure a smooth and safe docking process. Furthermore, the collision avoidance constraints are strictly enforced, ensuring that the tanker remains within the designated operational zone without colliding with the lab walls or the target satellite throughout the maneuver.

Fig. 6 shows the tanker's trajectory, expressed in the RSW reference frame, where blue, purple, and red represent the initial, final, and intermediate tanker states, respectively. It can be seen in this figure that the tanker has a smooth change in position and orientation and docks with the target successfully while no collision happens. Also, the tanker remains entirely inside the admissible zone during the docking phase. It is noteworthy that due to the relative motion dynamics in orbit, since approaching the target satellite from the left-hand side (the space between the target satellite and the Earth) requires less energy than its right-hand side, MPC has chosen this trajectory for docking.

The time histories of the reference position are shown in Fig. 7, where it is illustrated that it reaches the desired docking point,

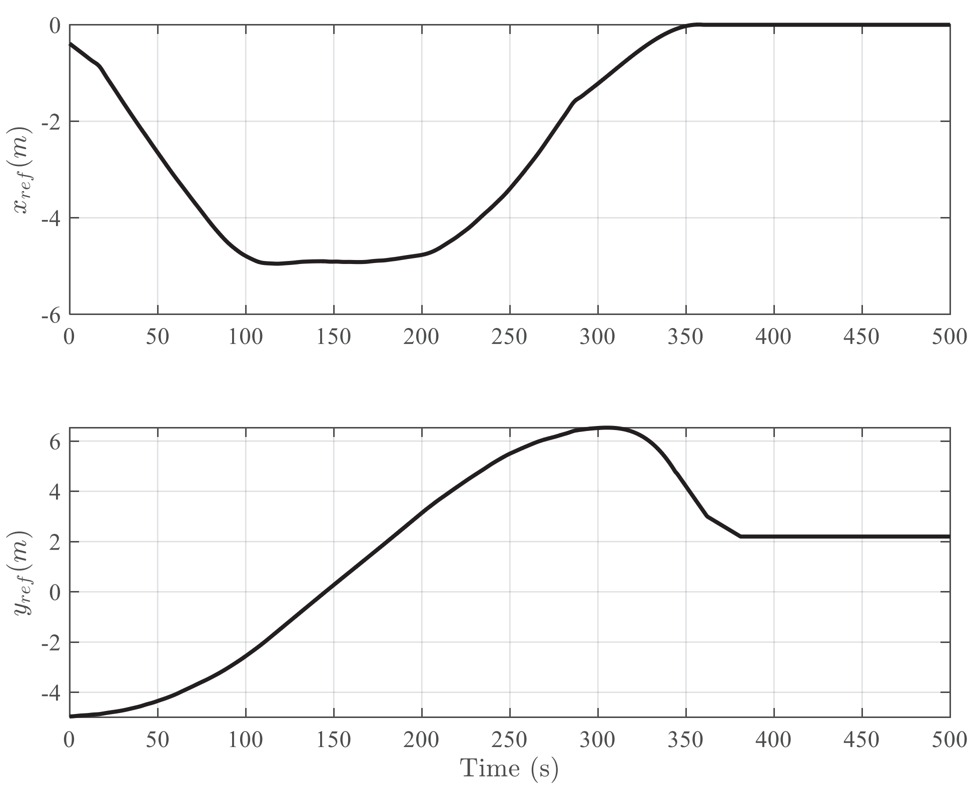

Fig. 7. The time histories of the reference trajectory of the tanker designed by MPC.

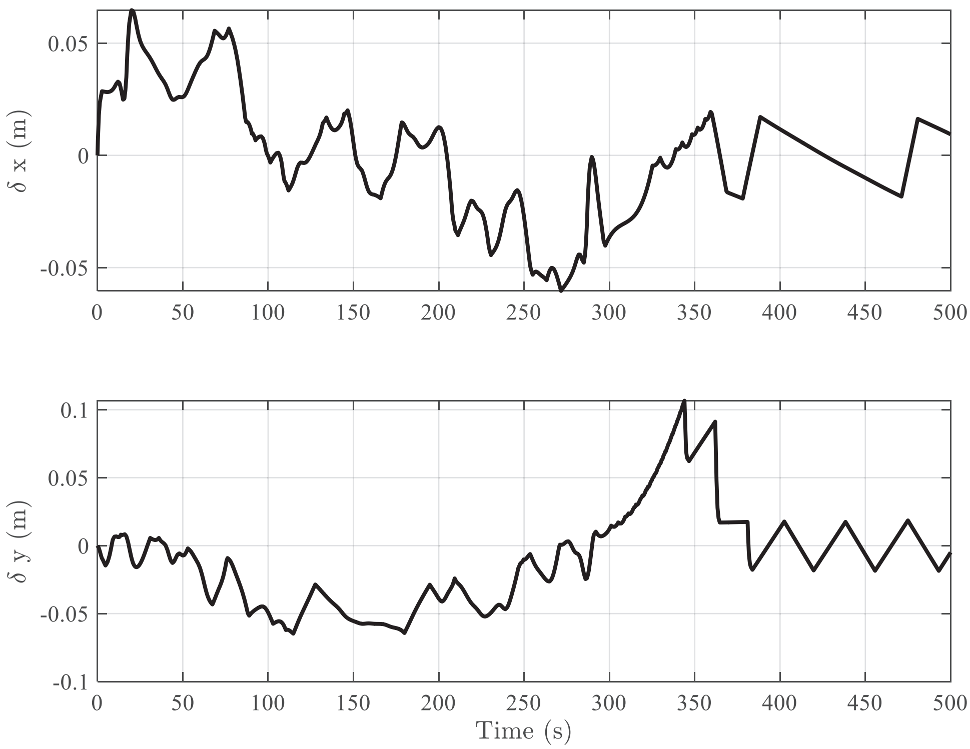

Fig. 8. The error of controlled position with respect to the reference trajectory.

and the docking state is maintained. Moreover, it is visible that the tanker moves just toward the y direction during the last twenty seconds before the docking instance, which guarantees accurate and safe docking. Fig. 8 shows the error of the tanker position with respect to the reference trajectory. It is obvious that the proposed controller can drive the tanker to track the reference trajectory successfully. The fluctuation visible in the position error is due to the discontinuous nature of the thrusters and fuel sloshing. Moreover, the effects of sloshing and thrusters are visible in the time history of the tanker velocity, Fig. 9. Similarly, the attitude of the tanker is controlled as commanded, as shown in Fig. 10. However, the effect of thrusters is entirely visible in the angular velocity behavior.

In Fig. 11, the variation of the angle of the free surface of the fuel, θ, over time is shown in two cases: 1) with the estimation of θ and 2) without the estimation of θ. It can be seen that although the fuel sloshing motion is successfully suppressed in both cases, the settling time when θ is estimated is much shorter than when θ is not estimated by the Kalman filter. The effect of undamped sloshing on the tanker velocity is visible in Fig. 12, showing that the velocity in the first case is just affected by the thrusters. In contrast, in the second case, fuel sloshing causes a fluctuating behavior in the tanker velocity that might reduce the docking safety.

Fig. 13 shows the time histories of the thrusters' activity, where it can be seen that they are off most of the time, and as a result, the fuel consumption is minimal due to the optimal trajectory design using the MPC's optimizer. It is worth mentioning that the impulsive nature of the thrusters can cause fuel-sloshing excitation. Thus, the lower the number of impulses, the lower the fuel-sloshing.

It should be noted that while the numerical simulations are conducted for a circular orbit, the proposed MPC-LQG framework is adaptable to elliptical and other non-circular orbits. The gravitational force (f_g) is calculated in real-time based on the satellite's position and orbital parameters, ensuring accurate control inputs for the thrusters. This flexibility allows the system to maintain reliable performance across different orbital regimes without structural modifications, enhancing its applicability to various on-orbit refueling missions.

5.3. Comparative performance analysis: MPC-LQG vs. SMC

To further evaluate the performance of the proposed MPC-LQG control framework, a comparative analysis was conducted against Sliding Mode Control (SMC), a widely used robust control technique [67]. Unlike the proposed approach, SMC does not incorporate collision avoidance constraints, limiting its applicability in scenarios where safety is a primary concern. To ensure a fair comparison, the initial points for the SMC simulations were selected within the admissible zone around the target satellite, ensuring that the tanker would not collide with the target during the docking maneuver.

The comparison was based on four key performance indicators: docking point velocity, which represents the root mean square (RMS) velocity of the tanker at the final docking point and is crucial for ensuring a safe and smooth docking process; docking accuracy, which measures the RMS deviation from the desired docking point, ensuring that both positive and negative errors are appropriately accounted for; control effort, representing the RMS value of the cumulative force applied by the thrusters throughout the docking maneuver, directly influencing fuel consumption; and average fluid oscillation, quantifying the RMS amplitude of sloshing inside the fuel tank, which impacts system stability during docking.

The results presented in Table 2 represent the average outcomes of 500 Monte Carlo simulation runs, each conducted with randomly perturbed initial conditions within the predefined admissible zone. The control parameters for the SMC approach were tuned according to the guidelines provided in [68] to ensure optimal performance.

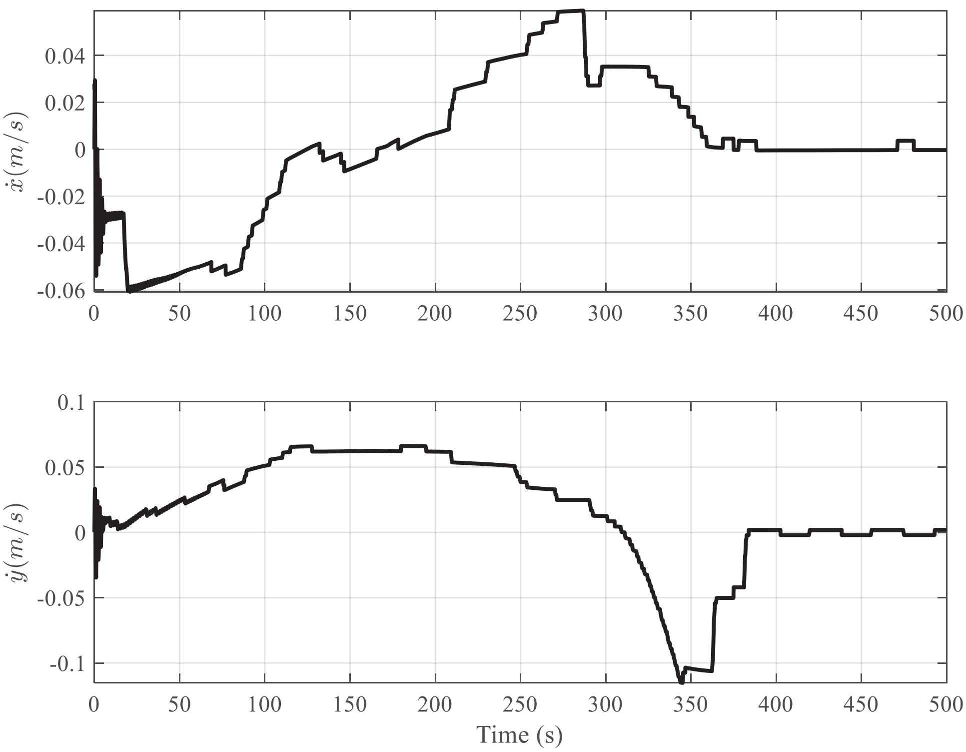

Fig. 9. The time histories of the tanker velocity with respect to the target satellite.

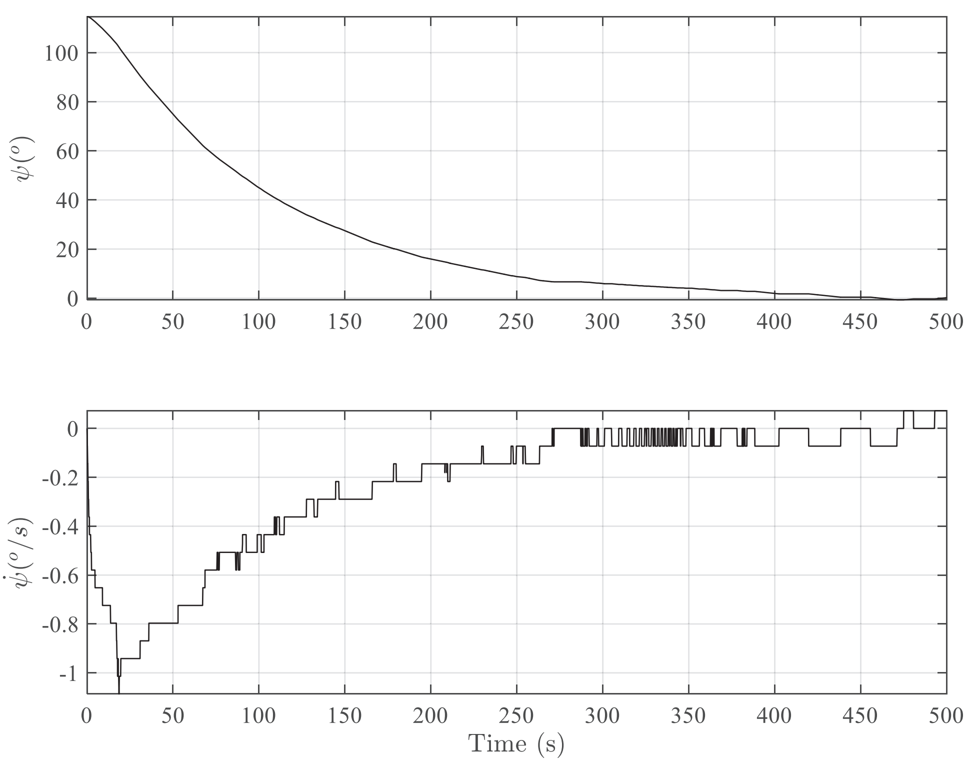

Fig. 10. The time histories of tanker orientation and the angular velocity.

The simulation results demonstrate that the proposed MPC-LQG framework outperforms SMC across all key performance metrics. Regarding docking point velocity, the MPC-LQG approach achieves a lower final velocity, ensuring a smoother and safer docking process. This advantage stems from the optimal reference trajectory generated by MPC, which incorporates a smooth docking constraint, gradually reducing the velocity as the tanker approaches the target. In contrast, the higher docking velocity observed with SMC is partially due to the inability to estimate and control fluid oscillations, as this method lacks an observer to monitor the fuel state. Consequently, uncontrolled fluid oscillations contribute to fluctuations in velocity, resulting in less precise control during the

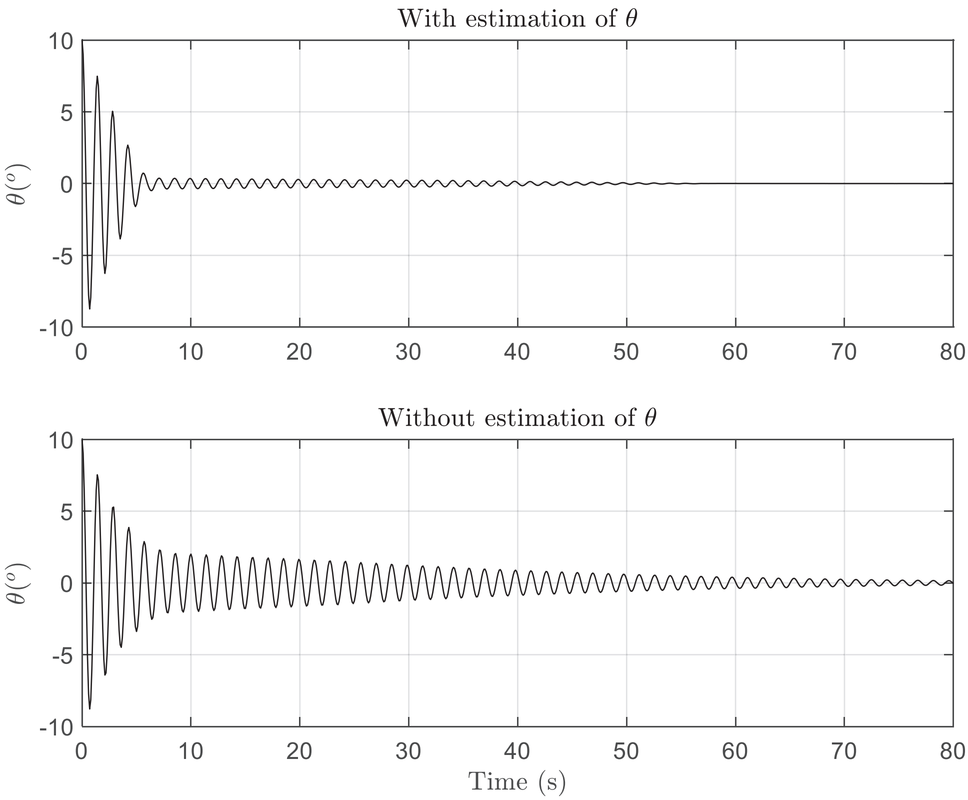

Fig. 11. The effect of estimation of θ on the sloshing control performance.

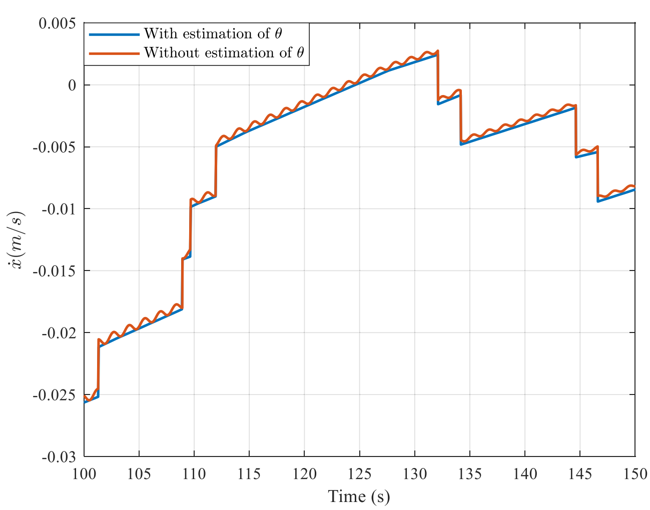

Fig. 12. The effect of estimation of θ on the tanker velocity.

Table 2
Comparative performance of MPC-LQG and SMC.

Performance Metric	MPC-LQG	SMC
Docking Point Velocity (m/s, RMS)	0.01	0.05
Docking Accuracy (m, RMS)	0.03	0.07
Control Effort (N·s, RMS)	8.2	13.5
Average Fluid Oscillation (rad, RMS)	0.01	0.08

final docking phase. Additionally, the chattering behavior inherent to SMC further exacerbates fluid oscillations during the maneuver, increasing velocity variations as the tanker approaches the docking point.

In terms of docking accuracy, the MPC-LQG framework also demonstrates superior performance. By following an optimal reference trajectory generated by MPC, the system can maintain a more precise approach toward the target, minimizing positional deviations at the final docking point. Conversely, SMC exhibits higher docking errors, partly due to the dead zone introduced to mitigate chattering effects during the final docking phase. While reducing the size of this dead zone could potentially improve docking accuracy, it would also result in increased control effort and more pronounced fluid oscillations, negatively impacting both fuel efficiency and system stability.

Regarding control effort, the MPC-LQG framework demonstrates greater efficiency, requiring less cumulative thruster force compared to SMC. This advantage is attributed to MPC's ability to account for orbital dynamics when generating the reference trajectory, allowing the system to optimize control inputs and minimize fuel consumption throughout the docking process. In contrast, SMC relies on a more reactive control approach that does not incorporate predictive optimization, leading to higher thruster usage and greater fuel consumption.

Finally, the average fluid oscillation experienced during the docking maneuver is significantly lower with the MPC-LQG approach. By continuously estimating the fluid state using the LQG observer and applying corrective control actions, the proposed framework effectively suppresses sloshing disturbances, maintaining system stability throughout the maneuver. In contrast, SMC lacks the ability to estimate and compensate for fluid oscillations, resulting in more pronounced fuel motion that further exacerbates chattering effects, leading to higher overall oscillation amplitudes.

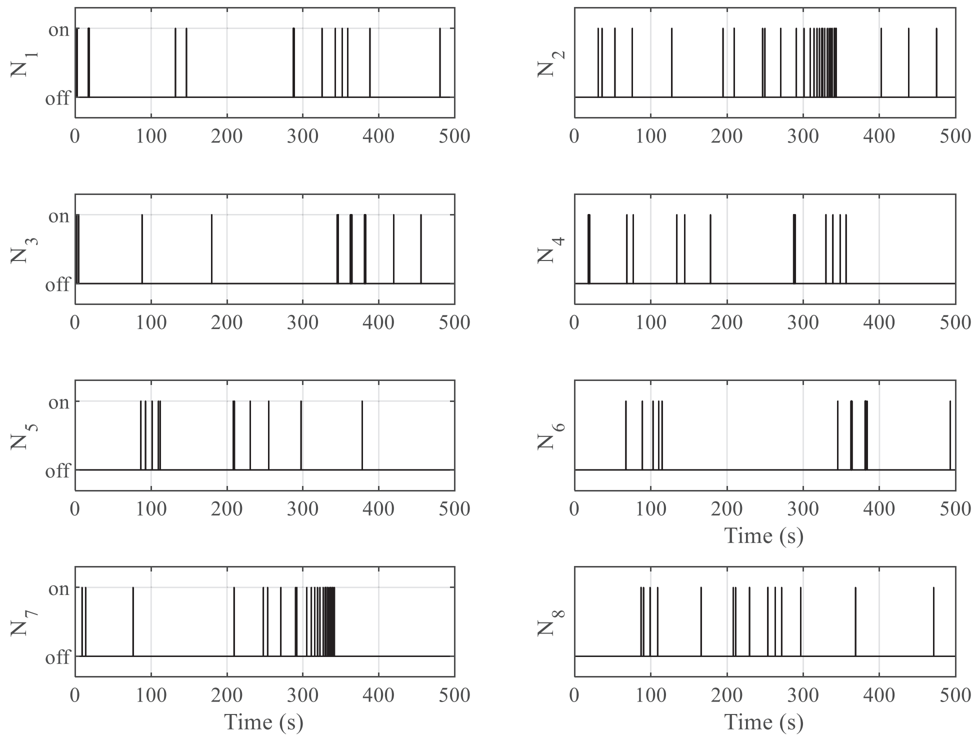

Fig. 13. The time histories of the thrusters firing over the docking phase.

6. Conclusion

In conclusion, this paper presents a control strategy for on-orbit satellite refueling, explicitly focusing on the docking phase. On-orbit servicing, including refueling, is a critical technology for maintaining and extending the lifespan of satellites in space. However, docking control is challenging due to the uncertain and dynamic nature of the fuel sloshing disturbance. A combination of model predictive control and linear quadratic gaussian control was proposed to address this issue. The fuel sloshing dynamics in the tanker were modeled using a spherical pendulum. The model predictive control algorithm generated a reference docking trajectory that satisfies collision avoidance constraints and optimizes fuel consumption. The linear quadratic gaussian control was then employed to track the reference trajectory and suppress fuel sloshing motion. Based on the capabilities of the Zero-G Lab facilities of the University of Luxembourg, numerical simulations were performed to evaluate the effectiveness and feasibility of the proposed approach. The results demonstrate that the proposed control strategy is capable of achieving a safe and fuel-efficient docking trajectory in the presence of fuel sloshing disturbance. The performance comparison between the proposed MPC-LQG framework and the sliding mode control method demonstrates the superiority of the proposed integrated approach in terms of control effort, docking accuracy, and safety. The results show that the MPC-LQG framework achieves more fuel-efficient control with a lower cumulative control effort due to its ability to optimize thruster usage while accounting for orbital dynamics. Additionally, the docking accuracy is significantly improved due to the optimal reference trajectory generated by MPC and the precise state estimation provided by LQG, allowing the system to maintain a more accurate and stable approach toward the target satellite. Moreover, the safety of the docking maneuver is enhanced through the collision avoidance constraints incorporated within the MPC optimization, ensuring that the tanker maintains a safe distance from the target satellite and remains within the designated operational zone of the Zero-G Lab. These improvements collectively demonstrate the effectiveness and reliability of the proposed method, making it a more robust and practical solution for on-orbit refueling missions. As part of future work, it is suggested to conduct experimental testing of the proposed scenario using the floating platforms available in the Zero-G Lab facilities. This will serve to validate the numerical results and optimize the control parameters for improved performance. Furthermore, it is recommended to investigate the system's performance in terms of disturbance rejection, considering factors such as gravity gradient, third body effects, and J_2 disturbances. Additionally, studying the impact of system uncertainties, including thruster misalignment, thruster failure, and valve dynamics behavior, would be beneficial for a comprehensive analysis of the system's behavior and robustness.

This study employs a spherical pendulum model to represent the fuel sloshing dynamics within the satellite's fuel tank. While this model effectively captures the primary sloshing effects and is computationally efficient, it has limitations when applied to non-spherical tanks (e.g., cylindrical or irregular geometries) commonly used in modern satellites. The simplified nature of the spherical pendulum model may not fully capture the complex fluid behavior that can occur in non-spherical tanks, particularly during rapid maneuvers or under microgravity conditions. To address these limitations, future research should explore advanced sloshing models that better represent real-world fuel behavior. Potential extensions include multi-mode pendulum models, which can capture higher-order sloshing modes, as well as finite element simulations and computational fluid dynamics (CFD) techniques that offer more accurate and detailed representations of fluid-structure interactions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

M. Amin Alandihallaj: Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Andreas M. Hein:** Supervision, Writing – review & editing. **Jan Thoemel:** Conceptualization, Supervision, Writing – review & editing.

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